

Economics

Higher level

Paper 3

SME Predicted Paper – Advanced

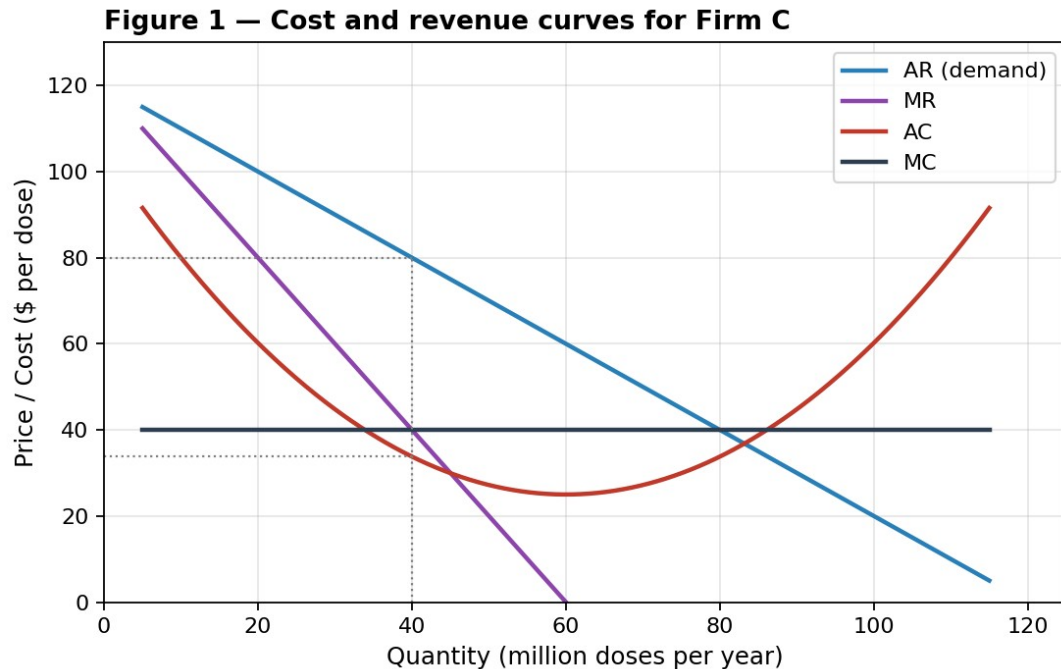
1 hour

Instructions to candidates

- You are permitted access to a calculator for this paper.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers must be given exactly or correct to two decimal places.
- You must show all your working.
- Use fully labelled diagrams where appropriate.
- The maximum mark for this examination paper is [60 marks].

1. Pharmaceutical markets for patented drugs often exhibit monopoly characteristics. Firm C holds a 20-year patent on a branded medicine used to treat a chronic disease.

Figure 1 shows Firm C's cost and revenue curves in the market for this drug. Firm C faces constant marginal cost of production.



(a) (i) Define the term **monopoly**.

[2]

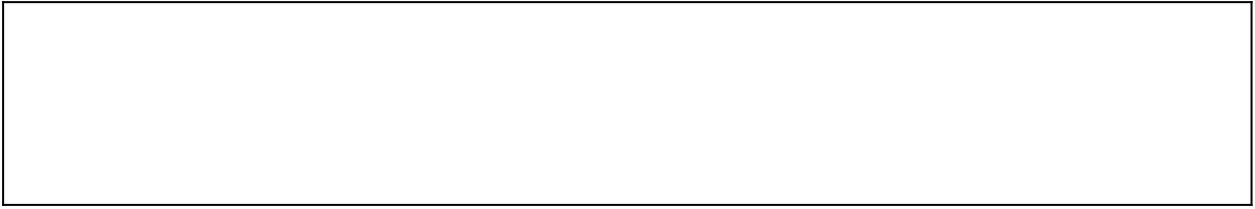
(ii) Using **Figure 1**, identify Firm C's profit maximising level of output.

[1]

(iii) Using **Figure 1**, calculate Firm C's abnormal profit at the level of output identified in part (a) (ii).

[2]

(iv) With reference to **Figure 1**, identify the allocatively efficient level of output for this market. [2]



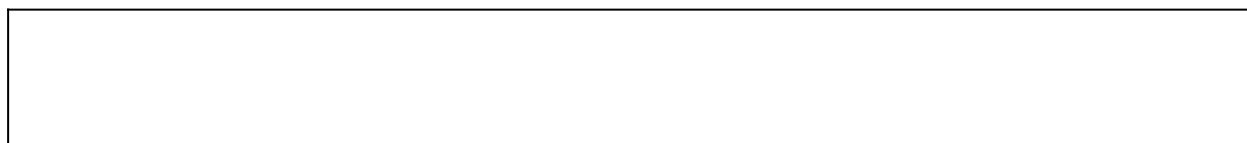
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(v) Using a price-discrimination diagram, explain how Firm C can increase total revenue by charging different prices in markets with different price elasticities of demand. [4]



Many pharmaceutical inputs (such as plant-derived active ingredients) are drawn from common pool resources. Access to these resources is not easily excludable but their use reduces availability for others.

(vi) State **one** of the characteristics of common pool (common-access) resources. [1]



Firm C engages in international third-degree price discrimination. In 2023 it sold 22 million doses in high-income markets at \$120 per dose and 45 million doses in low-income markets at \$18 per dose.

(vii) Using the information above, calculate Firm C's total revenue from the two markets combined in 2023. [2]

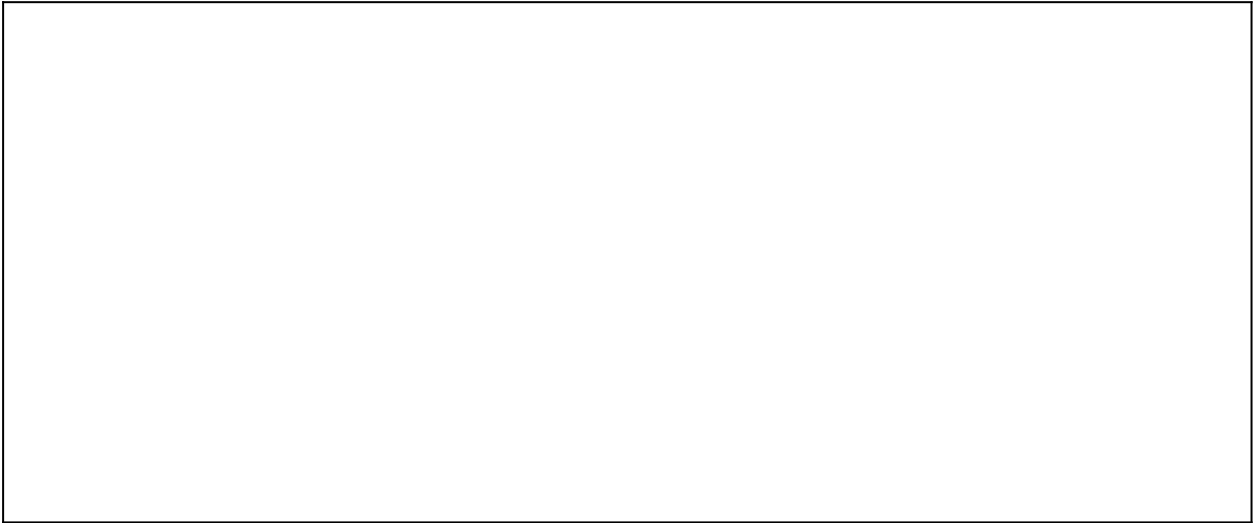


Overall, the patented-pharmaceuticals market faces well-documented welfare and sustainability challenges:

- Monopoly pricing reduces allocative efficiency and restricts access to essential medicines, particularly in low-income countries.
- High prices can deter adherence to treatment, leading to poorer public-health outcomes and higher long-term healthcare costs.
- Concentration of R&D in high-margin diseases leaves neglected tropical diseases under-researched.

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(viii) Using **Figure 1** and a welfare-analysis diagram, explain how monopoly pricing generates a welfare loss (deadweight loss) compared to the allocatively efficient outcome. [4]

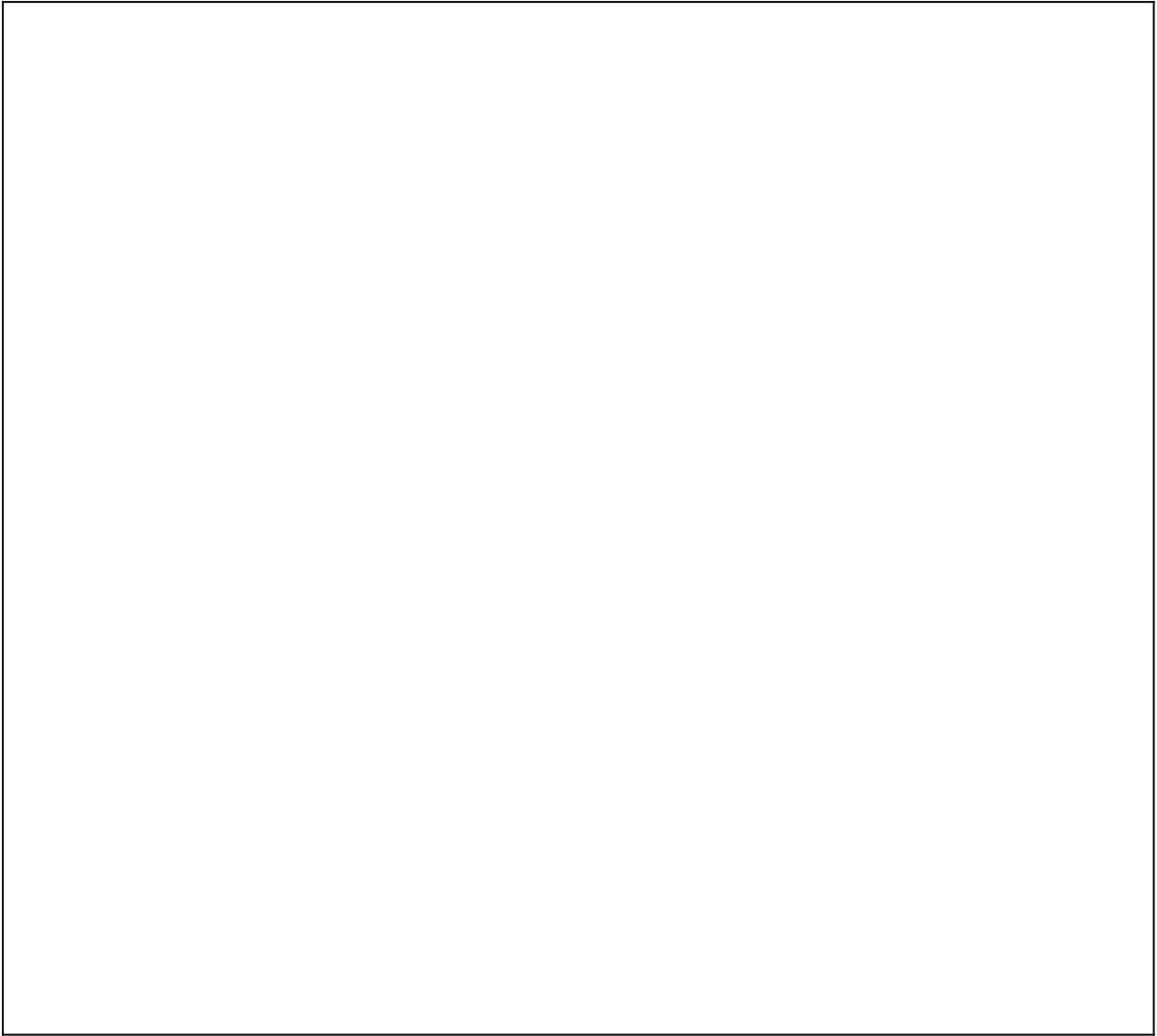


The US government imposes a per-unit excise tax of \$5 per dose on Firm C's domestic sales in 2023 as part of a price-regulation initiative. Assume Firm C sold 22 million doses in the US high-income market in 2023.

(ix) Using the information above, calculate the total indirect tax revenue collected by the US government in 2023. [2]



(b) Using the text/data provided and your knowledge of economics, recommend a policy that could be introduced to improve access to essential medicines while preserving incentives to innovate. [10]



2. Argentina is a large upper-middle-income Latin American economy that has experienced repeated episodes of high inflation, currency crises and sovereign debt problems. Table 1 below provides selected macroeconomic data.

	Nominal GDP (tn ARS)	GDP Deflator	Real GDP (tn ARS)	CPI (YoY %)	Policy rate (%)
2022	72.0	100.0	72.0	94.8	75.0
2023	165.0	210.0		211.4	133.0

Table 1: Selected macroeconomic data for Argentina, 2022–2023

(a) (i) Using the information in **Table 1**, calculate Argentina's real GDP in 2023 (to 2 dp). [2]

(ii) Using the information in **Table 1**, calculate the percentage change in real GDP between 2022 and 2023 (to 2 dp). [2]

Inflation in Argentina accelerated sharply during 2022 and 2023, reaching triple digits and threatening to tip into hyperinflation.

(iii) Define the term **hyperinflation**. [2]

(iv) Outline **one** cause of the very high inflation rate observed in Argentina. [2]

The Argentine peso (ARS) depreciated sharply against the US dollar over 2022–2023, from around ARS 130/US\$ to ARS 800/US\$ (official rate). A parallel 'blue-dollar' rate traded even weaker.

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(v) Using an exchange-rate diagram, explain **two** possible reasons for the depreciation of the peso. [4]

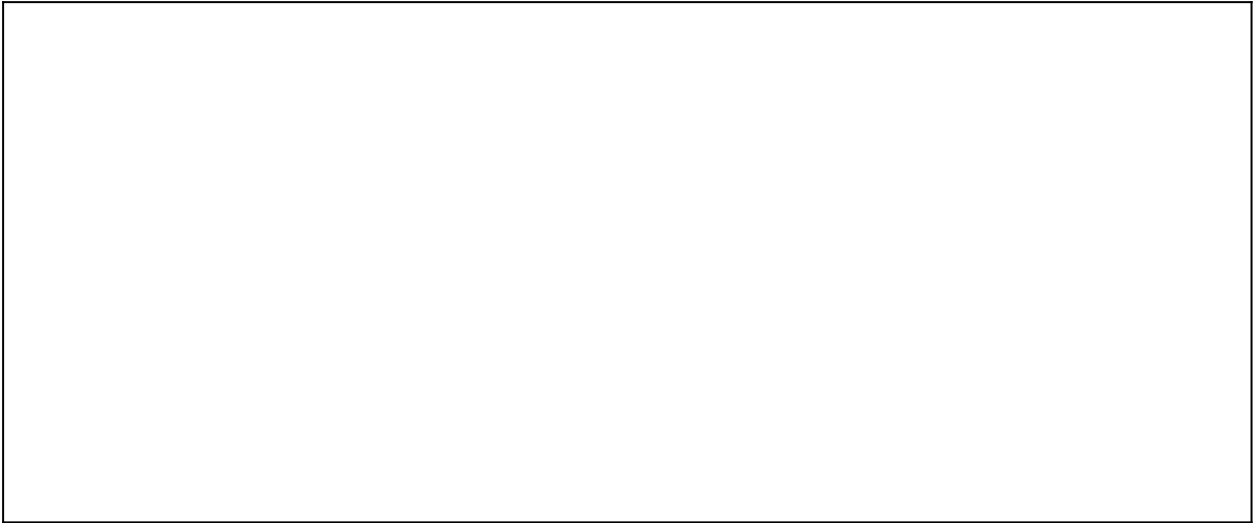
Table 2 below provides selected balance-of-payments data for Argentina in 2023 (US\$ billion).

Item	\$ billion
Exports of goods	66.8
Imports of goods	73.7
Exports of services	14.2
Imports of services	20.6
Net primary income	-17.2
Net secondary income	0.9

Table 2: Balance-of-payments data for Argentina, 2023

(vi) Using **Table 2**, calculate the current account balance for Argentina in 2023. [2]

(vii) Using an AD/AS diagram, explain **two** implications of a sharp currency depreciation for Argentina's price level and real output. [4]



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In late 2023 Argentina announced plans including drastic fiscal spending cuts, removal of price controls, and discussions around full dollarisation of the economy. Over the last decade, Argentina's real GDP growth has averaged below 1%, highlighting the vulnerabilities of an economy with persistent macroeconomic instability.

(viii) Outline why persistent macroeconomic instability can pose a barrier to economic growth and/or economic development. [2]

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(b) Using the text/data provided and your knowledge of economics, recommend a policy mix that the Argentine government could use to simultaneously address high inflation and the current account deficit. [10]

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